
BIF401_Bio-informatics

Current Final Term Questions

Full solved By M Zaman

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Improvements Of Chou Fasman Algorithm??

Special consideration for:

- Nucleation regions
- Membrane proteins
- Hydrophobic domains
- Consider variable coil and loop sizes besides the from tetra peptide turns
- Consider local protein folding environments
- Solvent accessibility of residues
- Protein structural class
- Protein's organism

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Intra-Molecular Distance Algorithms?

- Proteins are considered as rigid bodies.
- They are placed in a 3D Cartesian coordinate system.
- Structural alignment in 3D.
- E.g. VAST, LOCK

Inter-Molecular Distance Algorithms?

- Proteins are considered as rigid bodies.
- They are placed in 2D.
- Structural alignment using internal distances and angles.
- E.g. DALI, and SSAP.

Fourth relationships in Jacobson Algorithm

The matrix is filled by four different positions

1. Left
2. Bottom
3. Diagonal
4. Left/Bottom elements

Expasy

It is developed by **Swiss Bioinformatics Institute (SIB)**. Website provides access to databases and tools Proteomics, genomics, phylogeny, systems biology, population genetics, transcriptomics etc. can be searched. <http://www.expasy.org/>

Q. How loops are found after the finding of alpha and beta sheets?

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Tetra peptide Formula??

Ans:

After computing the propensity of alpha helices and beta sheets, we need to settle for loops.

- Let's see how can we find out the loops using Chou Fasman Algorithm.
- For any j^{th} residue in sequence, we calculate $f(\text{Total}) = f(j) f(j+1) f(j+2) f(j+3)$ (tetra peptide).

If

- $f(\text{Total}) > 0.000075$.
- The average value for $P(\text{turn}) > 1.00$ in the tetra peptide
- The averages for the tetra peptide are such $P(\text{a-helix}) < P(\text{turn}) > P(\text{b-sheet})$
- It is a Turn!

Beta Sheets

Beta Sheets are another secondary structure that can be formed as a result of hydrogen bonding between the protein back bones. Some amino acids have a preference for making Beta Sheets. Typically, 5 to 10 residues are needed to make a Beta Strand

1. Parallel beta sheet.
2. Anti-parallel beta sheet

Beta Sheets have different Properties:

- Beta Strand

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- Beta Sheet
- Beta Barrel
- Beta Sandwiches

Secondary Structures of RNA

A simple nucleotide sequences of RNA is called as Primary structure and denoted by 1' while when these nucleotides fold together and form a complex structure that is called secondary structure and denoted by 2'. The preferred structure of RNA is 2' which has many structural patterns like **Helices, Loops, Bulges and Junctions**

Secondary Structures of Proteins -II

2' structures or secondary protein structures are formed as a result of H-Bond formation between N and C terminus in a protein backbone. Types of 2' structures include Alpha helices and Beta sheets.

Protein Fragmentation & Techniques?

We compare the experimental mass with theoretical data base mass of protein and on base of closeness we rank or score it. If several proteins have same score than selection is done by using another technique protein fragmentation.

We fragment the protein or peptide and ionize it, it helps us to measure the fragment masses as the same ways as their precursor.

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There are different **techniques** for protein fragmentation.

- Electron Capture Dissociation (ECD)
- Electron Transfer Dissociation (ETD)
- Collision Induced Dissociated (CID)

Need For Chou Fasman Algorithms?

Chou Fasman Algorithm helps to predict Alpha Helices, Beta Sheets and Turns. The algorithm is based on statistical occurrence of Amino Acids in known structures. The relative frequencies of each amino acid based on known protein structures solved with X-ray crystallography. From these frequencies a set of probability parameter were derived.

Ab Initio Modelling With its LIMITATIONS

Ab initio methods have Anfinsen's thermodynamic hypothesis at the center. These methods attempt to identify the structure with minimum free energy

Limitation

- Computationally expensive
- Suitable for proteins with less than 100 residues

PDB file Format

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HEADER – Contains a brief description of the structure, the date and the PDB ID code.

TITLE – The title of the structure.

COMPND – Brief details of the structure.

SOURCE – Identifies which organism the structure came from.

KEYWDS – Lists a set of useful words/phrases that describe the structure.

AUTHOR – The scientists depositing the structure.

REVDAT – The date of the last revision.

Difference B/T Uni_Prot & Swiss_Prot

Both UniProt and SwissProt are the online database for proteins.

Swiss-Prot contains human curated protein information

- Accession number, unique identifier
- The sequence
- Molecular mass
- Observed and predicted modifications

Protein sequences from various species and organisms can be found in uniprot. SwissProt is the manually annotated version of the UniProt Database.

How we can Score Each Proteins? Strategies??

We can score each protein in the way that it get maximum score and low quality matches should get low scores.

After filtering the multiple charges we get the only the peaks having charge 1. And after this filter we compare it with protein data base.

Strategies

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1. X-Ray Crystallography
2. Nuclear magnetic resonance (NMR)

How many conditions are involved in finalizing region as beta sheet??

3 out of 5 amino acids have $P(\beta) > 1.0$. That region is declared as beta sheet. Extend beta sheet to both sides until 4 contiguous residues average $P(\beta) < 1.0$

That is declared end of the beta sheet. Those regions are finalized as beta-sheets which have average $P(\beta) > 1.05$ and the average $P(\beta) > P(\alpha)$ for that region.

Amino Acid Classifications into How many Groups?? Their Names

1. HYDROPHILIC AMINO ACIDS

Since H and C introduce very little dipole moments in hydrophobic amino acids, these amino acids are non-polar. Hydrophobic amino acids are mostly found at the inside of folded proteins. Hydrophilic group contain the chain of C and H group in their R group. E.x alanine, valine, leucine, etc.

2. POLAR AMINO ACID

POLAR AMINO ACID These amino acids are polar but are not charged i.e. no net charge on the amino acid. Prefer to reside / interact with aqueous environments. Mostly found at the surface of folded proteins. E.x serine, asparagine, glutamine and cysteine.

Amino acids Characteristics:

- Polarity
- Hydrophobicity
- Charge states

Homology modeling 7 Steps:

1. Template recognition and initial alignment
2. Alignment correction
3. Backbone generation
4. Loop modeling
5. Side-chain modeling
6. Model optimization
7. Model validation

Three different strategies for structure prediction

1. Homology Modelling
2. Threading/Fold Recognition
3. Ab Initio Modelling

Limitations of Homology Modelling

- Large Bias towards structure of template
- Cannot study conformational changes
- Cannot elicit new catalytic/binding sites

BLOSSUM Matrices

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BLOSUM matrices can be used to align the protein sequences. BLOSUM matrices was first purposed in 1992 by Henikoff et al. BLOSUM matrices is also called the Block substitution matrix without any gap although it has mismatches in sequences.

There are three steps to compute the BLOUSM Matrices.

Step 1: Eliminate sequences that are identical in x% positions

Step 2: Compute observed frequency $f_{i,j}$ of aligned pair A_i to A_j . Hence, $f_{i,j}$ becomes the probability of aligning A_i and A_j in the selected blocks.

Step 3: Compute f_i which is the frequency of observing A_i in the entire block.

In Silico fragment scoring

- Count the matches between in silico and in vitro peaks.
- Give an equivalent score to the candidate protein.
- Weigh each of the fore mentioned match by the mass error.
- Accumulate the score

RAW File Format & MASS Spectrometer

Mass spectrometer outputs data with ionic mass/charge ratios & respective ion intensities. RAW file is a format in which an instrument outputs data in binary form.

Multiple RAW file formats are prevailing in the industry. Each vendor has its own unique RAW file format. You can convert proprietary formats into open formats

Dot plots cannot capture insertions, deletions and gap indications. How can we deal with them?

To find matching in nucleotides and amino acids of two sequences we use dot plot method. But dot plot cannot capture the insertions,

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deletions and gaps in the sequences. To deal with this situation we modify the dot plot. We represent the matching nucleotides with +1 while gaps, substitutions, insertions and mutations can be represented as -1 in dot plot

Nussinov -Jacobson Algorithm

Nussinov-Jacobson (NJ) Algorithm is a Dynamic Programming (DP) strategy to predict optimal RNA 2' structures, Proposed in 1980. Computes 2' structures with most nucleotide coupling.

A scoring matrix is initialized to record scores in NJ Algorithm .For filling scoring matrix, the maximum score from 4 matrix positions is chosen.

PDB

PDB contains protein structure information. It has the coordinates of C-Alphas for over 50,000 proteins. Protein structures can be visualized using this information.

Main points to be focused in N-J Algorithm

- Scoring Matrix
- Matrix Initialization
- Scoring method
- The 4 different positions to be considered for calculating matrix, Left, Bottom, Diagonal, and Left/Bottom elements.

Needleman Wunsch Algorithm

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Needleman Wunsch Algorithm is the way for alignment. The method is same like dot plot but it computes the scores in different way. We Start with a zero in the second row, second column. Move through the cells row by row, calculating the score for each cell.

MGF File Format

MGF – Mascot Generic Format. MGF is a simple human-readable format for MS/MS data developed by Matrix Science. Mascot Search Engine available at this link online.
http://www.matrixscience.com/search_form_select.html

What Are Uses Of Mass Spectrometry?

Mass spectrometry (MS) is an analytical technique that ionizes chemical species and sorts the ions based on their mass-to-charge ratio. In simpler terms, a mass spectrum measures the masses within a sample. Mass spectrometry is used in many different fields and is applied to pure samples as well as complex mixtures.

What Happened When Ph Is Less Than The Pk Value?

If the pH is higher than the PK, then the compound will be deprotonated. A further consideration is the charge on the compound. Acids are neutral when protonated and negatively charged (ionized) when deprotonated. Bases are neutral when deprotonated and positively charged (ionized) when protonated.

Pros and Cons of Ab initio method?

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Pros: Ab Initio methods can fold any target sequence using only physical atomic properties. Predictions are mostly accurate and correctly describe the natural folding process.

Cons: Ab initio methods are the very difficult to design (energy function). These methods are slow due to the huge possibilities.

How OH of RNA make it less stable than DNA?

The two hydroxyl groups make RNA less stable than DNA because it is more prone to hydrolysis Hence, RNA has a short life span and is more prone to degradation.

Pairs of nucleotides of DNA to form helix

DNA Double helix DNA molecule is double helix structure contain base pairs composed of nucleotides and these nucleotides are composed of sugar phosphate group and are bind with each other with hydrogen bonds.

Adenine “A” pairs with Cytosine “C”, While Thymine “T” binds with Guanine “G”

Normally all the nucleotides are same in both DNA and RNA except one position in RNA which is U (Uracil) and in DNA it is T (Thiamin)

Scoring Peptide Sequence Tags (PST) + Formula

According to scoring scheme if a candidate protein matches ‘n’ PSTs, then its score can be given by:

$$PST_{Score} = \sum_{i=0}^n Length(PST_i)^2$$

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Additionally, if we include RMSE to the scoring system, then it can highlight better PST matches. And **RMSE is the root mean square error.**

Tandem mass spectrometry (Tandem MS)

Tandem mass spectrometry (MS-MS) is a related technology in which compounds are separated by molecular weight by one mass spectrometer, fragmented as they exit, and identified on the basis of their fragments by a second mass spectrometer

N-J Algorithm Scoring Matrix

A scoring matrix is initialized to record scores in NJ Algorithm .For filling scoring matrix, the maximum score from 4 matrix positions is chosen.

The matrix is filled by four different positions. Left, Bottom, Diagonal, and Left/Bottom elements. In this way all complementary nucleotides coupling is catered.

Advantages and Disadvantages of Threading

Advantages:

Threading helps predict secondary structures of proteins towards tertiary structure prediction

✓ For the “Twilight Zone” with low alignment quality and identity, threading is useful.

Disadvantages:

Novel proteins cannot be predicted using threading Fewer than 30% of the predicted first hits are true remote homologues

✓ Validation of each result is necessary

Types of secondary structures of RNA:

1. Single stranded: 3' end may fold on to the 5' end.
2. Helices: Double stranded RNA helix of stacked base pairs

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3. Hairpin loop: The loop of the hairpin must at least four bases long to avoid steric hindrance with base-pairing in the stem part of the structure
4. Bulge Loops: Bulges, are formed when a double-stranded region cannot form base pairs perfectly.
5. Interior loop: Interior loops are formed by an asymmetric number of unpaired bases on each side of the loop.
6. Junctions or intersections: Junctions include two or more double-stranded regions converging to form a closed structure. The unpaired bases appear as a bulge

Fasta and blast difference

- FASTA can perform quick comparison of protein and nucleotide sequence
- Can also perform genome and proteome similarities search
- Its also available online like BLAST. FASTA can search sequence databases and identify unknown sequences by comparing them to the known sequence databases. This can help obtain information on the parent organism, function and evolutionary history

Blast

BLAST for Basic Local Alignment Search Tool is an algorithm for comparing primary biological sequence information, such as the amino acid sequences of proteins or the nucleotides of DNA sequences.

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A BLAST search enables a researcher to compare a query sequence with a library or database of sequences, and identify library sequences that resemble the query sequence above a certain threshold.

Protein Structure Prediction

Complex protein structures enable proteins to perform complex functions. We know over a million protein sequences but only about 100,000 protein structures. Estimating exact protein structures is very difficult. It's difficult to crystallize proteins. Even if we manage to get protein's X-Ray, to reconstruct the structure is extremely complex.

Since we know so many sequences, they can be used for predicting protein structures. This indeed is possible and helpful.

What is hydrophobicity of amino acid?

In chemistry, hydrophobicity is the physical property of a molecule that is seemingly repelled from a mass of water. In contrast, hydrophiles are attracted to water.

Hydrophobic molecules tend to be nonpolar and, thus, prefer other neutral molecules and nonpolar solvents.

Purines & Pyrimidine

Adenine and Guanine collectively called Purines while Cytosine, Uracil, and Thymine are called as Pyrimidine.

Translation

Cells are built of proteins and carbohydrates and these proteins are made in results of transformation of RNA molecule and this transformation is called as translation.

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Translation takes place in ribosome of cell and ribosomes after reading the information of mRNA collect the amino acids from cell cytosol which is the part of the cytoplasm that is not held by any of the organelles in the cell.

Transcription

Transcription All cells are made of carbohydrates and proteins and for these cells DNA codes the information which makes the RNA and protein both.

Difference b/w DNA & RNA?? What are Nucleotides??

If we talk about the composition of DNA and RNA molecule than these are composed of four other molecules which are named as Nucleotides.

These molecules are Adenine (A), Cytosine (C), Thymine (T), Uracil (U), and Guanine (G).

DNA molecule although is double stranded and RNA is single stranded but there is difference in sugar composition.

RNA has Ribose sugar and DNA has de-oxyribose sugar: Difference between RNA and DNA sugar RNA Adenine and Guanine collectively called Purines while Cytosine, Uracil, and Thymine are called as Pyrimidine. when phosphate, nitrogen base and sugar come together if there is (OH) than molecule is RNA and if there is (H) in sugar than molecule is DNA

How we can determine the structure of unknown protein by using homology modeling?

Protein sequence gives rise to its structure

If another protein which has a similar sequence also has its structure known, the structure of an unknown protein can be predicted based on that similar protein So, it is then possible to identify unknown protein structures by just examining the homologous protein sequences

Conclusions

- Sequence Identity • Alignment Length

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What is the purpose of 4th relationship in Nussinov Jacobson's algorithm?

Aligning Mutations in Sequences

Removal and addition of amino acids in proteins and nucleotides in DNA, RNA by using Gaps named as Indels.

Mutation is totally different from Indels, because in Mutation we replace the amino acid with other amino acids and replace the nucleotides with other and we don't use Gaps is inserted in template or target for mutation.

MSA

MSA involves progressive alignment of sequences. Doing so many progressive alignments can be slow. CLUSTALW is an online tool to perform MSA.

Zuker's Algorithm & Martinez Algorithm

Zuker's Algorithm helps us to compute the stabilizing energies (-ve) and also destabilizing energies (+ve values). And also compute the sum of +ve and -ve energies.

Martinez Algorithm is improvement on it. Making combination of all possible structures is time consuming, Martinez Algorithm favors those 2' structure which are energetically more feasible.

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Types of FASTA

There are six types of FASTS:

- fasts35 Compare unordered peptides to a protein sequence database
- fastm35 Compare ordered peptides (or short DNA sequences) to a protein (DNA) sequence database
- Fasta35 Scan a protein or DNA sequence library for similar sequences
- Fastx35 Compare a translated DNA sequence (6 ORFs) to a protein sequence database
- tfastx35 Compare a protein sequence to a DNA sequence database (6 ORFs)
- fasty35 Compare a DNA sequence (6ORFs) to a protein sequence database

Identity & Similarity

There are two concepts for sequence analysis

1. Identity
2. Similarity

Identity means the counting number of nucleotides or amino acids which exactly match when two biological sequences are matched.

Formula for Identity: $\text{Identity} = \frac{\text{No. of Matches}}{\text{smaller length}} \times 100$

Similarity means the comparison between two different sequences calculated by alignment approach.

Protein structure strategies

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3. X-Ray Crystallography
4. Nuclear magnetic resonance (NMR)

We need to know the structure of the protein to calculate atomic distances. Atomic distances tell us about atomic interactions with neighboring atoms. To determine the structure, we use XRay or NMR.

Types of forces involved in folding

- Hydrophobic interactions
- Electrostatic interactions
- Hydrogen bond
- van der Waals

Phylogenetic tree

All molecules have an evolutionary history. Phylogenetics is the science of studying evolutionary relationships. Phylogenetics has led to the creation of relationship trees between various species of Bacteria, Archaea, and Eukaryota.

Types of Phylogenetic Trees

Scaled Trees • Branch lengths are equal to the magnitude of change in the nodes

Unscaled Trees • Only representing the relationship between sequences

Nussinov-Jacobson (NJ) Algorithm

Nussinov-Jacobson (NJ) Algorithm is a Dynamic Programming (DP) strategy to predict optimal RNA 2' structures, Proposed in 1980. Computes 2' structures with most nucleotide coupling.

HOW IT WORKS Steps:

- Create a matrix with RNA sequences on top and right
- Set diagonal & lower tri-diagonal to zero
- Start filling each empty position in matrix by choosing the maximum of 4 scores

Domains

Domains are semi-independent functional structures in a protein. Protein may contain multiple domains. Hence, we can try to classify proteins by their domains.

Locally Compact – Domains interact (H-bonds) more internally than externally. Domains have a hydrophobic core. Domains are contiguous (min. chain breaks). Domains have a minimal contact with rest of the peptide. Solvent area in contact with each domain should not vary significantly upon separating two separate domains.

Write the Types of Domains ?

Types of Domains

- Alpha Domains
- Beta Domains
- Alpha/Beta Domains
- Alpha + Beta Domains
- Alpha & Beta Multi-Domains
- Membrane & cell-surface proteins

Difference B/W Peptide Mass & Shotgun Proteomics

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Shotgun Proteomics digest the whole protein and mix first and compared with database. And peptide mass finger printing involves in protein separation followed by single protein's peptide analysis.

Integrative Scoring Schemes

Three individual scores can be obtained:

1. MW Match score
2. PST Match score
3. In vitro & In silico Match score

Structural Classification (CATH)

- **Class** Similar secondary structure content All α , all β , alternating α/β etc.
- **Architecture** Also called FOLD Major structural similarity SSE's in similar arrangement
- **Topology** Super Family Probable common ancestry Family membership
- **Homology** Same Family Clear evolutionary relationship Pairwise sequence similarity > 30%

If you find Any Mistake Tell me as soon as possible.

Remember us in your prayers

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Admins of All About Zoology

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